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Consumers' Responses to Virtual Influencers as Advertising Endorsers: Novel and Effective or Uncanny and Deceiving?

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ABSTRACT

Virtual influencers, generated on the computer, are changing the advertising industry. More and more brands utilize virtual instead of human influencers as endorsers for their marketing campaigns for a variety of reasons, e.g., better control of the influencer's behavior, cost savings, or zeitgeist motives (desire to be "up to date"). However, there is currently little research on whether significant advertising goals are achieved using virtual influencers. We analyze, in an initial study, whether consumers find such influencers appealing compared to human influencers and whether they are able to identify that the influencer is not real. Our results show that consumers find it difficult to identify virtual influencers as such and that they still have more positive attitudes toward human endorsers in advertising campaigns. However, virtual endorsers can lead to higher perceived ad novelty. In a second study, we further examine whether the advertised product category functions as a moderator. Results show that perceived congruence between virtual influencers and product is dependent on the product category and leads to higher advertising effectiveness. Our studies lead to the implication that marketers should carefully consider the use of virtual influencers in accordance with the aspired advertising goals and brand values.

The Emergence of a Novel Type of Endorser

Novel technologies, especially artificial intelligence (AI), are already prevalent in different industries and are now becoming increasingly important for the advertising landscape (Qin and Jiang 2019; Rodgers 2021). In light of this, a new phenomenon of "nonhuman alternatives to traditional celebrity endorsers" is finding its way into brand communication (Thomas and Fowler 2021, 11). It is now possible for marketers to digitally create "virtual influencers" according to a desired image and to use them for advertising and social media activities. Advertising expenditure in the field of influencer marketing is expected to reach $4.14 billion in the US in 2022 (up from US$2.42 billion in 2019), a growing proportion of which will make use of this new type of virtual endorser (eMarketer 2021).

The use of virtual characters for commercial purposes is not entirely new. In the music industry, for example, virtual representations like Hatsune Miku, the anthropomorphized mascot of a synthesized voice ("vocaloid") enjoy remarkable success. She is considered a "virtual celebrity" and even performed as an opening act for Lady Gaga's world tour in 2014, where she appeared as a projected hologram. Through advances in computer technology (CGI) it is now possible to create virtual influencers with such distinct, human-like features that it can be hard to identify them as not real. Covid-19 further accelerated their "rise" as brands focused more on creative digital solutions—while most humans were stuck at home, virtual influencers could still make appearances in the outside world (Conti, Gathani, and Tricomi 2022).

Some virtual influencers are given fully developed fictional lives with elaborate personalities and they display human emotions, such as heartbreak and affection, on social-media sites. Consumers are not always clear about the origin of the created content, particularly whether it is controlled by human or machine intelligence (da Silva Oliveira and Chimenti 2021; Robinson 2020). Experts predict a significant rise of virtual-

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influencer prominence in the upcoming years (Appel et al. 2020). Various types of virtual influencers have been created, for example Shudu, an African influencer who endorses fashion and cosmetic brands such as Fenty Beauty (Rosenstein 2018). Other examples range from the more cartoon-like figure Noonoouri, who claims to be vegan and cooperates with different luxury brands, to the ultra-realistic Japanese character Imma, who sports pink hair and collaborates with IKEA and Dior (Travers 2020a). However, the virtual influencer currently gaining the most attention (3.1 million followers on Instagram to date) is Lil Miquela, a computer-generated image "identifying" as a half Brazilian, half Spanish female from Los Angeles (Moustakas et al. 2020). Not only does she promote various brands, but she also acts as a supporter of social causes (Black 2019). Her Instagram activities seem so lifelike and close to what human influencers display on social media that many of her followers appear to struggle with the idea of her not being real (Drenten and Brooks 2020). In 2018, Lil Miquela was even named as one of the 25 most influential people on the internet by *Time* Magazine (Time 2018). A large number of mainstream (e.g., KFC) and luxury brands (e.g., Calvin Klein, Prada, and Chanel) have already partnered with virtual influencers to promote new product lines or have selected one as their brand representative (Robinson 2020).

But why do established brands rely on virtual rather than real human beings to endorse their products? There seem to be compelling arguments, such as the ability to design the virtual influencer in perfect alignment with the brand's personality (Moustakas et al. 2020). Another decisive factor is the attention that a virtual influencer can draw to the brand, owing to its still-existing novelty factor, especially when addressing a younger target group (Forsey 2019). Furthermore, there is the hope of future cost savings and a lower risk that the influencers will engage in scandals and thus harm the brand image (Dempsey 2018; Thomas and Fowler 2021)—as long as the influencer is not misused by individuals not acting in the interest of the brand, for example by leaking incriminating content.

However, there has been very little research either on how consumers respond to virtual influencers or on the comparison between virtual and human influencers. The present research aims to fill these gaps, answering several calls from marketing and advertising literature. In a first step, we analyze how virtual influencers are evaluated by consumers compared to human influencers. Besides attitude toward the ad, we consider perceived ad novelty. We then examine whether the use of virtual influencers is able to influence brand perception. This is in response to recent requests from various researchers (Appel et al. 2020; Thomas and Fowler 2021) to pay more attention to the perception of virtual characters and particularly concerning differences between virtual and real endorsers in their effect on consumers or brands. Last but not least, we approach the topic from a consumer-protection perspective by raising the question whether disclosure about their digital nature has an impact on the perception of the influencers. Labeling virtual influencers as digital creations is demanded by consumer protectionists for transparency reasons (Callahan 2021).

In a second step, we ask whether the perception of virtual influencers depends on the product category being advertised. We claim that virtual influencers (which are marked as such) may be more apt for certain types of products. Virtual influencers are technical masterpieces, so we assume that they are a better suit for technical products (e.g., smartphones, smart speakers) than for products that are not technology related (e.g., personal care products). This is especially relevant because virtual endorsers are predominantly found in the fashion and cosmetics domain, where known brands either cooperate with virtual influencers or even create their own fictive characters (Naumann and Daubenbüchel 2021). We thus respond to Thomas and Fowler (2021), who were among the first to investigate virtual or AI influencers empirically in an advertising context. In their discussion, they raise the question whether the effectiveness of the endorser type is dependent on the category of the advertised product. We use match-up hypothesis as the theoretical foundation for the assumption that congruence between virtual influencer and advertised product is important for advertising evaluation (e.g., Kamins 1990; Wright 2016).

To summarize, our research contributes to a better understanding of this novel type of virtual endorser (in comparison to a "real" endorser) and its effectiveness in advertising, as well as with respect to different product categories. To the best of our knowledge, there have been no experimental studies in this regard. Scholars as well as practitioners would therefore benefit from more insights into modes of virtual endorser effectiveness and possible applications.

Theoretical Background

Definition of Virtual Influencers in Marketing

When trying to establish a definition of a virtual influencer, we start with the term *virtual*. By virtual

we understand something that is created by technology and appears to exist but does not, at least not in the physical world (Cambridge Dictionary 2022). This contrasts with anthropomorphized robots, which do exist in real life where people can touch them. Virtual influencers, however, belong to the broader category of virtual humans; a generic term for human-like characters that "exhibit human-like behaviors … and might also show other human characteristics, such as emotions (or) empathy" (Burden and Savin-Baden 2019, 3). Virtual humans used in marketing can further be categorized into different subgroups depending on their specific area of use, comparable to real human beings with different "jobs."

The first subgroup of virtual humans is virtual assistants, which are, like chatbots, used to perform service tasks. While chatbots' primary mode of interaction is through messaging and they are mainly used to solve easy queries in customer support, virtual assistants are a more sophisticated version of a chatbot and can be used for more advanced and diverse tasks across different fields, such as educational tutors or personal sport coaches.

The second subgroup (the one we focus on in our studies) is virtual influencers that are used as endorsers for advertising and brand communication. According to Thomas and Fowler (2021, 12), a virtual influencer can be understood as "a digitally created artificial human, who is associated with Internet fame." This does not mean that they are necessarily known to the masses, but they are able to interact with followers on social media. In addition, they may feature fictional personalities, increasing their potential to interact. Moreover, if a virtual influencer "uses software and algorithms to perform tasks like humans," it can be further specified as an AI influencer (p. 12).

There are also other forms of virtual humans, such as virtual celebrities (e.g., the aforementioned Hatsune Miku); however, in this article we focus on those that are used by brands for particular purposes.

Literature Review on Virtual Influencers and Virtual Humans

To date, there are only a handful of scientific articles examining the impact of virtual influencers in advertising. There is, for example, Robinson (2020), who raises in a discussion paper some important ontological and ethical issues concerning virtual influencers. Moustakas et al. (2020) use a qualitative approach to explore the potential advantages and disadvantages of using virtual influencers for advertising purposes. By questioning experts (not consumers), they have identified the main points of potential and challenge. While these findings are an important first step in the understanding of this new phenomenon, there is still little quantitative empirical knowledge on consumers' approval of the technology. To the best of our knowledge, there is only one article empirically determining consumer reactions to virtual influencers, in particular AI influencers.

Thomas and Fowler (2021) conducted two important studies to determine the impact of a transgression made by either an AI or a human influencer. In both transgression conditions, the influencer made a politically incorrect statement on social media. Across both studies, the results show a decrease in the perceived value of both the influencer and the brand, owing to the wrongdoing. The authors (2021, 20) state that AI influencers are evaluated "in a manner similar to the employment of a novel celebrity endorser." However, when trying to diminish the transgression's harm, it is better to replace the AI influencer with a human influencer. The underlying explanation (p. 20) is that "AI influencers are more likely than celebrity endorsers to be viewed as similar and interchangeable and, thus, are also seen as more likely to commit comparable transgressions in the future." Thomas and Fowler's (2021) studies show that there are certain similarities but also significant differences in the perception and evaluation of human vs. AI influencers, contributing substantially to the understanding of this new phenomenon.

However, there are issues still to be addressed in order to find out if, and under which conditions, virtual influencers are suitable endorsers compared to human ones. While Thomas and Fowler (2021) focused on a case of misconduct, we do not investigate transgressions but rather analyze "normal" (typical) positive advertising messages presented by virtual influencers. In addition, in Thomas and Fowler's studies, respondents were confronted with a news article about an endorsement deal between a brand and the AI influencer. In our studies, we create an advertisement focusing on the visual depiction of the respective influencers. We also consider different product categories, taking into account the importance of fit between endorser and brand (Choi and Rifon 2012; Till and Busler 2000). What is more, participants in Thomas and Fowler's (2021) studies were informed about the influencer's artificial nature. We investigate the role of disclosing the virtual influencer's "digital

origin," checking whether there are differences in the evaluation of influencer uncanniness if labeled or not.

The starting point of our investigation is the currently unanswered question of whether consumers have a positive or negative attitude toward virtual influencers compared to human ones with respect to positive advertising messages. There is a lot of uncertainty regarding approval of virtual influencers, with the notion that consumers tend to be rather skeptical. As explained above, Thomas and Fowler (2021) demonstrated that people are more likely to forgive human mistakes than those of AI figures. This could be owing to a phenomenon called algorithm aversion: individuals tend to have a general antipathy toward algorithms, particularly in the case of algorithms making mistakes (Dietvorst, Simmons, and Massey 2018). Dietvorst, Simmons, and Massey (2015) showed that people lose confidence in algorithm forecasts more quickly than in human forecasts, even if they see the algorithm outperform the human forecaster. This means that respondents would prefer to rely on an underperforming human than on an algorithm, especially regarding tasks that are typically performed by humans (Castelo, Bos, and Lehmann 2019).

These findings are not limited to algorithm research: Lou, Kang, and Tse (2022) found that participants rated an online customer service experience better when they interacted with a human employee instead of a virtual assistant, demonstrating that human users naturally are fonder of services offered by other human beings. Longoni, Bonezzi, and Morewedge (2019) discovered a serious reluctance in participants to accept medical care delivered by AI providers vs. human providers. This obstacle to AI adoption is explained by Schmitt (2020, 4) by an "inclination to prefer humans over AI because of a fundamental bias toward their own kind." Among the scarce literature on virtual influencers, experts in a qualitative study expressed serious concerns about the ability of virtual influencers to develop trust in consumers, since they are not able to try out the products and give genuine recommendations (Moustakas et al. 2020). This could negatively affect sustained interest in the advertising brand, since, especially in the field of brand communication, there is agreement on the point that authenticity, trust, and "realness" are essential values for a brand (e.g., Portal, Abratt, and Bendixen 2019).

Therefore, we assume that consumers have a more positive attitude toward a human over a virtual endorser, resulting in a more positive attitude toward the human influencer ad compared to the virtual influencer ad. Thus, we formulate the following hypothesis:

> Hypothesis 1: When advertising with a human influencer (vs. a virtual influencer), attitude toward the influencer will be evaluated more positively (a), which in turn increases general attitude toward the ad (b).

On the other hand, experts suggest that the use of virtual influencers presents some opportunities. Tayenaka (2020) states that the reason for brands' affinity toward virtual influencers is their "freshness." Moustakas et al. (2020) confirm this and also identify the novelty factor as the key success driver. Yap (2018) explains further that, in these times, where countless "ordinary" human influencers are flooding the industry, the phenomenon of virtual influencers stands out from the crowd, providing some diversity. Particularly in brand communication, novelty and uniqueness of the created content/advertisement is a very important issue since brand differentiation can lead to enhanced brand attractiveness and value (Aaker 1990; Gupta et al. 2020). Therefore, we assume that an advertisement featuring a virtual influencer will lead to a higher perceived novelty of the advertisement and may also spill over to a higher perceived innovativeness of the advertising brand. Thus, we hypothesize:

> Hypothesis 2: When advertising with a virtual influencer (versus a human influencer), perceived novelty of the ad will be evaluated more positively (a), which in turn increases perceived innovativeness of the brand (b).

In addition, we approach the topic from a consumer-protection point of view, since there are currently no clear regulations that oblige companies to disclose the "non-humanness" when advertising with virtual influencers (Barnett 2021). Consumer protectionists are clamoring for strict regulations to force virtual-influencer creators to mark social-media posts or advertisements with a clear indication that the influencer is not a human being but a computer-generated entity (Callahan 2021). It is argued that, without such labeling, consumers can be misled into believing that a virtual influencer's product recommendations are based on their autonomous experience and assessment (Masteralexis, McKelvey, and Statz 2021). Callahan (2021) also points out the counterargument that when looking closely everybody should be able to identify the computer-generated nature of the influencers. However, some virtual influencers appear so lifelike that it can be hard to distinguish them from actual human beings (Powers 2019).

To bring more light into the debate about whether labeling of virtual influencers is worthwhile, we investigate in the first study whether consumers are able to identify a virtual influencer as such if it is not labeled. Moreover, we analyze whether labeling might also change the evaluation of the virtual influencer. The extreme human likeness of a virtual influencer could potentially evoke a feeling of eeriness, also known as the phenomenon of the uncanny valley (Mori, MacDorman, and Kageki 2012). Recently, Rosenthal-von der Pütten et al. (2019) confirmed the uncanny-valley effect using a functional magnetic resonance imaging (FMRI) study, which revealed that, when viewing the more or less human-like figures, significant changes were observed in the ventromedial prefrontal cortex (VMPFC), which is believed to reflect sympathy. The more the figures resembled humans, the more the activity of the VMPFC increased, but when the subjects were confronted with the extremely human-like being, the activity decreased.

The uncanny-valley effect can also be evoked through the inability to ascertain whether an entity is virtual or human. This category uncertainty is provoked whenever an entity shifts between one category and another (MacDorman and Chattopadhyay 2016). When individuals struggle to determine the nature of an ambiguous entity, negative evaluations are likely to be evoked (Yamada, Kawabe, and Ihaya 2013). Not knowing or not being able to decide whether a figure is real or virtual can evoke perceived discomfort. Carrying this over to virtual influencers, we assume that an extremely lifelike look of a virtual influencer will lead to a higher feeling of uncanniness if the virtual influencer is not labeled as such. Thus, we hypothesize:

> Hypothesis 3: If a virtual influencer is *not* labeled as such, individuals perceive a stronger feeling of uncanniness toward the influencer than if the virtual influencer is labeled.

Study 1

Design and Stimulus Development

Study 1 employed a 2 (type of endorser: virtual influencer versus human influencer) × 2 (labeling: absent versus present) between-subjects design. The stimulus showed an advertisement presenting the influencer as the new face of the brand Lancôme (see Figure 1). Both endorsers received the fictive name Emily Woods.

We chose the brand Lancôme since it is one of the most prestigious beauty brands globally and is well known throughout different age groups (Bladt 2019). The brand identity is distinct, and it is widely known that the brand stands for caring and decorative cosmetics. Lancôme is also renowned for the frequent use of influential celebrities to endorse its products, which was important to ensure that the advertisement was perceived to be realistic. In recent years it has been focusing more on social-media marketing and has added influencers to its set of brand ambassadors, such as Taylor Hill or Dove Cameron, thus appealing more to younger generations (Greenberg 2016). All these activities result in the brand being well known among both older and younger target groups.

We selected a virtual influencer (Lil Miquela) and a human influencer (Emily Bador), who look very similar. According to some sources, Lil Miquela is considered to be the digital version of British influencer Emily Bador (Dodgson 2019). Additionally, in the two labeling conditions, participants received brief background information on the influencer's origin. In the human-endorser condition, respondents were informed that the influencer was born in New York, while in the virtual condition they received the information that the influencer was created on the computer of a graphic designer in New York.

Confounding Check

A pretest was conducted to determine whether the advertising design and the influencers' looks and overall style were perceived to be similar. $N = 29$ respondents were exposed to both advertisements and asked to evaluate similarity of ad design ($M = 5.70$), influencer look ($M = 5.26$) and influencer style ($M = 5.15$) on 7-point scales, without telling the respondents that one of the influencers is not "real." Because the means are relatively high, we can expect comparability of the two ads.

Sample

We collected data online on $N = 352$ respondents (M age $= 28.83 \pm 10.40$), who were randomly assigned to the four experimental conditions. Respondents were recruited by 24 students of a German university master's course in market research. The recruiters were each instructed to recruit at least eight respondents (with the option to collect more), and to choose only female participants because we considered that male participants would lack a feeling of involvement for a

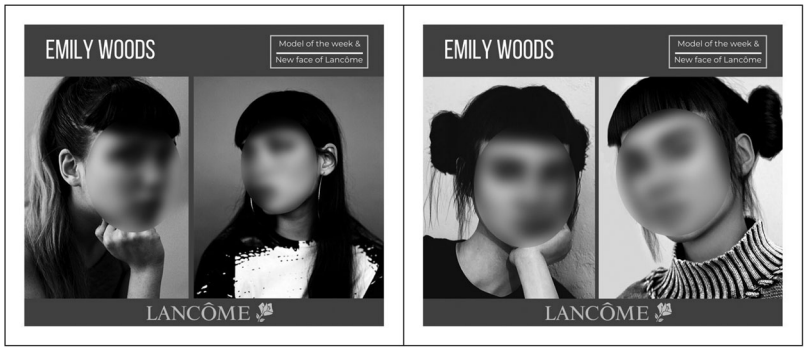


Figure 1. Stimuli used in Study 1 (blurred for copyright reasons).

Lancôme (cosmetic brand) advertisement. The online survey was administered through Qualtrics and sent out by the recruiters via email or social media. In exchange, recruiters received course credits. We excluded 14 respondents because they either interrupted the survey or exceeded the time limit (> 25 minutes). In all four conditions, we asked if the influencer that was shown in the advertising (Emily Bador or Lil Miquela) was already known to the individual and excluded seven respondents (five from the virtual, two from the human-influencer conditions) to avoid prior endorser knowledge as a potential confounding variable in our studies (mere exposure effect). We also had a control question, asking respondents to indicate the brand visible in the advertisement by a multiple-choice question (three options: Dior, Chanel, Lancôme). Only two respondents, who were excluded anyway because of the time exceedance, gave a wrong answer to the question. Chi-squared tests and ANOVAs showed that the four groups (2×2 experimental design) did not differ significantly with respect to socioeconomic variables, involvement in cosmetic and care products, technology affinity, and technology aversion (all $ps > .36$).

Measures

All constructs in both studies were measured on 7-point scales. We measured attitude toward the endorser and the ad with four items adapted from Silvera and Austad (2004) anchored by 1 (*not pleasant, uninteresting, not likeable, bad*) and 7 (*pleasant, interesting, likeable, good*) and averaged ($\alpha_{att.mod} = .92$; $\alpha_{att.ad} = .89$). Perceived novelty of the advertisement was measured using five items adapted from Andrews and Smith (1996) anchored by 1 (*conventional, predictable, ordinary, commonplace, dull*) and 7 (*unconventional, novel, unique, original, exciting*) and averaged ($\alpha = .82$). We measured brand innovativeness with four items adapted from Pappu and Quester (2016), such as, "The brand Lancôme is able to use new technologies in the cosmetic industry," and, "The brand Lancôme is able to bring innovative products to the market." To measure uncanniness, we used five items (eerie, freaky, supernatural, uncanny, thrilling) of the uncanny-valley index developed and validated by Ho and MacDorman (2010) to complete the sentence, "The endorser seems to be …" and averaged them ($\alpha = .86$). We also considered control variables and included "product involvement" as a covariate, after checking the criteria for covariates suggested by Meyvis and Van Osselaer (2018). Product involvement was significantly correlated to the dependent variables attitude toward the ad ($r = .27$, $p < .01$) and perceived novelty of the ad ($r = .20$, $p < .01$) and thus used as covariate (Spilski, Gröppel-Klein, and Gierl 2018, 76).

Recognition Check

We included a control question in which participants were asked to remember whether the endorser in the advertisement was either virtual or a human being, to check for correct identification. In the groups where the endorsers were not labeled as real or virtual, we wanted to get an impression of whether the consumers were able to classify them correctly. In total, $n = 201$ (60.7%) respondents correctly indicated the "nature" of the endorser. To test for hypotheses 1 and 2, we included only those respondents who correctly identified the endorsers, otherwise the respondents either thought the human influencer was virtual or thought the virtual influencer was human (which could have biased our results). With respect to the question whether virtual influencers can be identified at all without labeling and the reactions that this triggers, we compared in a second step the participants in the virtual-influencer conditions who received the labeling versus not.

Table 1. Effect of endorser type on attitude toward the endorser and ad.

Descriptive statistics	Condition	*M*	SD	*n*
Attitude toward the endorser (mediator)	Virtual	4.41	1.22	105
	Human	5.39	1.19	96
Attitude toward the ad (dv)	Virtual	3.94	1.40	105
	Human	4.75	1.21	96

Mediation analysis	Unstandardized coefficient *b*	Standard error	*T*-value	*p*
Outcome: attitude toward the endorser				
Type of endorser (virtual vs. human)	.940	.170	5.538	< .001
Product category involvement (cov)	.120	.057	2.093	.038
Outcome: attitude toward the ad				
Attitude toward the endorser (med)	.700	.057	12.173	< .001
Type of endorser (Virtual vs. Human)	.087	.591	.591	.555
Product category involvement (cov)	.145	.047	3.092	.002

Indirect effects	Effect coefficient (Partially stand. effect size)	Bootstrapping standard error	Bootstrap CI: lower level	Bootstrap CI: upper level
Type of end. → att. end. → att. ad	.658 (.478)	.138	.411	.947

Results

Descriptive statistics (see Table 1) show that attitude toward the endorser was significantly ($p < .01$) lower in the virtual-endorser condition ($M = 4.41$; $SD = 1.22$) than in the human-endorser condition ($M = 5.39$; $SD = 1.19$). We used Hayes (2018) Process (v. 3.5, model 4) to conduct a mediation analysis with the type of endorser (virtual vs. human) as nominal scaled independent variable, attitude toward the endorser as mediator and attitude toward the ad as dependent variable. The results (see Table 1) indicate that advertising with a human endorser leads to a higher attitude toward the endorser, $b = 0.94$, $p < .01$, which in turn significantly increases attitude toward the ad, $b = 0.70$, $p < .01$. The indirect effect was significant, $b = 0.66$, bootstrapping 95% CI [0.411; 0.947]; zero is not included in the interval. There was no significant direct effect, $b = 0.09$, $p = .55$. These results support hypothesis 1 (a and b).

Hypothesis 2 was tested analogously, but this time we obtained significantly higher results with respect to perceived ad novelty for the virtual endorser ($M = 4.68$; $SD = 1.27$) than the human endorser ($M = 4.21$; $SD = 1.16$). Mediation analysis (Hayes 2018, model 4) showed that ad novelty is significantly higher when advertising with a virtual endorser, $b = 0.53$, $p < .01$, in turn leading to a higher perception of brand innovativeness, $b = 0.36$, $p < .01$. The indirect effect was significant, $b = 0.19$, bootstrapping 95% CI [0.363; 0.057]. We also found a significant direct effect, $b = 0.43$, $p = .04$, suggesting that there might be further mediators influencing the relationship between the type of endorser and the emergence of brand innovativeness (Hayes 2018). Hypotheses H2a and H2b can be confirmed. Descriptive statistics and the results of the mediation analysis are provided in Table 2.

To test for hypotheses 1 and 2, we only considered those respondents who correctly identified the endorsers. To answer the question as to whether respondents are able to identify a virtual influencer without labeling, we took a closer look at the two virtual endorser conditions and the number of respondents who correctly identified the virtual endorser. In the virtual-endorser condition without labeling, a lot of respondents classified the endorser incorrectly, thinking it was human instead of virtual (32%), or they could not tell (29%). In the labeled condition, only few respondents incorrectly remembered the endorser's origin (5%) or could not tell (3%). A chi-squared test confirmed that the proportion of respondents incorrectly classifying the endorser differs in the label versus no label condition ($\chi 2$ (1, $n = 165$) = 52.55, $p < .01$). We then checked whether there are also differences in the evaluation of endorser uncanniness depending on the labeling. The results indicated that the two virtual endorser conditions differ significantly regarding perceived uncanniness of the endorser, with higher ratings found for the non-labeled condition ($M_{\text{nolabel}} = 3.52$, $SD = 1.64$; $M_{\text{label}} = 2.92$, $SD = 1.30$; $t(103) = 2.0$, $p = .04$). So, the virtual endorser seems less uncanny when consumers are informed about its origin. Thus, H3 can be confirmed.

Discussion of Study 1

The results show that the human endorser outperforms the virtual with respect to attitude toward the

Table 2. Effects of endorser type on ad novelty and brand innovativeness.

Descriptive statistics	Condition	*M*	SD	*n*
Perceived novelty of the ad (mediator)	Virtual	4.68	1.27	105
	Human	4.21	1.16	96
Brand innovativeness (dv)	Virtual	4.70	1.44	105
	Human	4.15	1.55	96

Mediation analysis	Unstandardized coefficient *b*	Standard error	*T*-value	*p*
Outcome: perceived ad novelty				
Type of endorser (human vs. virtual)	.527	.169	3.111	.002
Product category involvement (cov)	.189	.057	3.296	.001
Outcome: brand innovativeness				
Perceived ad novelty (med)	.364	.084	4.326	< .001
Type of endorser human vs. virtual)	.425	.205	2.073	.040
Product category involvement (cov)	.125	.070	1.797	.074

Indirect effects	Effect coefficient (partially stand. effect size)	Bootstrapping standard error	Bootstrap CI: lower level	Bootstrap CI: upper level
Type of end. → novelty ad → brand inno.	.192 (.126)	.078	.363	.057

endorser and attitude toward the ad. Consumers' evaluation of the virtual influencer advertising was rather mediocre, with a mean of 3.94 in terms of attitude toward the ad. However, when it comes to novelty, the virtual endorser could "succeed." Perceived ad novelty was enhanced for the virtual influencer ad, which in turn led to an increase in perception of brand innovativeness. This suggests that the use of a virtual influencer is more strongly connected to the concept of innovativeness. We conclude that, while virtual-influencer liking was not particularly pronounced, their use may still be beneficial in terms of the novelty factor, since ad novelty was also able to positively alter the perception of brand innovativeness.

In this regard, it could be fruitful to take a closer look at other potential mediating variables between endorser type and brand innovativeness. Maybe the virtual influencer triggers a higher degree of fascination since it blurs the distinction between real and unreal. There are certainly some individuals who are fascinated by the intersection of real life and virtual reality—think, for example, of the real-life market for virtual products in the metaverse. This heightened fascination based on the blurring of lines might also improve brand innovativeness, a topic that could be investigated in a further study. Another variable that might signal innovativeness could be sustainability, particularly in light of the negative environmental impact of, for example, the (fast) fashion industry which is responsible for approximately 10% of global emissions (Dardana 2021). Working with virtual instead of human endorsers to showcase new collections, conduct virtual runway shows, or manage the purchase of digital clothing in the metaverse could improve the environmental balance of the fashion industry in the long term (Ionita 2022). Lee and Lin (2022) show that sustainability can strengthen the perceived innovative character of a product. In addition, the design of the virtual influencer itself could play a role; future studies could examine the relationship between virtual influencer design and novelty perception.

Apart from this, the question arises whether there are other product categories that would be better suited to virtual influencer advertising. In Study 1, we used a cosmetics brand, since currently virtual influencers most frequently partner with cosmetic and lifestyle brands (Moustakas et al. 2020). In Study 2, we analyzed whether product category functions as a moderator to explain virtual-endorser ad effectiveness based on the assumptions of the match-up hypothesis. We assume a better fit between virtual influencers and technical products (because they are innovative) than cosmetic products. With respect to the labeling debate, we stress the importance of disclosing the virtual nature of the endorser. Participants indeed had trouble identifying the influencer as virtual without a marking, often mistaking it for a human being. What is more, we found differences regarding the uncanniness perception of the two virtual endorsers; when labeled, the feeling of uncanniness was significantly reduced. This might be because consumers sense

something inexplicably "odd" about the virtual influencer, and the labeling may relieve the insecurity. However, uncanniness perception was not evaluated at the top end of the scale in either of the two virtual endorser conditions. With means of 2.92 and 3.52 on a 7-point scale, uncanniness seems not to be a huge problem for the virtual endorser.

Future studies could compare virtual influencers differing in their degree of visual realism to ascertain whether an even more realistic look would lead to a fall into the uncanny valley or if it could be overcome, as found in some 3D rendering studies (e.g., Seymour et al. 2021). Nevertheless, even if uncanniness is not rated particularly high, it is negatively correlated to attitude toward the ad ($r = -.274$, $p < .001$), proving its importance for advertising effectiveness. The fear expressed by virtual influencer supporters that labeling the models as "not real" could prove disadvantageous (Dahan 2020) is thereby invalidated. Thus, in Study 2 we added labeling information in both virtual endorser conditions.

Study 2

Theoretical Framework and Hypotheses

The effectiveness of an endorser may be dependent on the right combination of product and endorser: Kanungo and Pang (1973) were among the first to discover that the use of a particular endorser with a certain product can lead to positive evaluation of the product, while the same endorser can generate negative evaluations with another product. The so-called match-up hypothesis explains that there has to be a fit or match between product and endorser, which in turn influences ad- and product-related evaluations such as attitude toward the ad or purchase intentions (Kahle and Homer 1985; Kamins 1990; Till and Busler 2000; Wright 2016). The match-up hypothesis was tested based on different factors determining the match, such as attractiveness (e.g., Bower and Landreth 2001) and expertise (e.g., Till and Busler 2000). Initially, the emphasis of the match-up hypothesis was placed on the physical attractiveness of the endorser (Lee, Kim, and Kim 2014; see, for example, Kamins 1990; Levi, Varnali, and Tosun 2017). More recent studies have shifted the focus to perceived expertise or congruency as more promising match-up dimensions (Choi and Rifon 2012). For example, Schouten, Janssen, and Verspaget (2020) found a match between endorsers who are known for their sports achievements and products related to sports, namely an athlete (versus actor) endorsing a fitness bar (versus candy bar).

Literature remains unclear regarding the definitions of "match" and "fit" (Ang and Dubelaar 2006; Choi and Rifon 2012) but defines the term "congruence" as a state in which the relevant characteristics of the endorser are in line with the relevant characteristics of the product or brand (Misra and Beatty 1990). For example, a human endorser would be congruent with a cosmetic brand, since he/she could be considered an expert in that field (Fleck, Korchia, and Le Roy 2012; Till and Busler 2000). The question arises as to whether the same is true for a virtual endorser. Cosmetics such as lotions or perfumes need to be applied on a human body to fulfill their purpose. Because they are meant to affect a human's appearance, there is a close link between the relevant product characteristics and the advertising appeal "well-cared-for human face/body." Since the focus is more obviously on a human body, presenting cosmetics together with a virtual endorser implies a mismatch, because in such cases the advertising presenter (virtual influencer) is not able to display as a credible endorser for this category.

In contrast, we assume that products that are not dependent on a "real" human body, for example technical products, would be a better choice for virtual influencer advertisements. Characteristics relevant to technical products would be primarily innovativeness, something that virtual influencers also represent (as seen in study 1), or it is conceivable that they can handle technical tasks (as robots do). Innovative technical products—just like virtual influencers—represent a futuristic allure. Therefore, we assume that congruence between a virtual endorser and a technical product is higher than with a cosmetic product. For the human endorser we do not expect any differences depending on the product category. We hypothesize that there will be an interaction effect between type of endorser (human vs. virtual) and product category (cosmetic vs. technical product):

> Hypothesis 4: The virtual endorser will generate a higher congruence when combined with a technical product than with a cosmetic product.

Further, in line with previous studies on endorser/product congruence, we assume that a fit has significant effects on endorser and ad evaluations (Choi and Rifon 2012). When a positive attitude toward the ad is formed, purchase intention may also be positively affected—a relationship that has been examined many times.

Hypothesis 5: The higher congruence (hypothesis 4) will lead to higher perceived expertise of the endorser and in turn to an increase in attitude toward the ad. Attitude toward the ad will positively influence purchase intention.

Study 2: Empirical Results

Design and Stimulus Development

In study 2, we again employed a 2 (type of endorser: virtual versus human) × 2 (product category: cosmetic versus technical) between-subjects design. The stimulus showed an advertisement similar to the one in the first study, but this time using a concrete product in combination with a brand. We chose Samsung as the technical brand and Calvin Klein as the cosmetic brand because both of these brands have already partnered with virtual influencers (Petrarca 2019; Travers 2020b), to ensure a high degree of external validity. As products, we selected a body lotion and a smart speaker (see Figure 2 for the stimuli used in study 2). In a brief introduction, respondents were informed that they would see an advertisement for a smart speaker/body lotion endorsed by the virtual/human influencer Emily Woods (fictive name). This was followed by a disclosure about the endorser's origin. We stated that the human influencer was born in New York and that the virtual influencer was created on the computer of a graphic designer in New York.

Pretests

We conducted two pretests. First, we tested the underlying assumption that the relevant product characteristics of the technical item differ from those of the cosmetic product. Whereas the technical product consists of technical components and should be innovative, the cosmetic product can only "function" when applied on a body. $N = 52$ respondents (M age $= 30.42 \pm 12.00$) were randomly exposed to either the cosmetic or the technical product and asked to indicate which characteristics the product addresses. Six items measured these characteristics (7-point disagree/agree rating scales), including three items representing the technical characteristics ("The product is innovative," "The product consists of technical components," and, "The product is a highly technical product") and three items representing the cosmetic characteristics ("The product is applied on a body," "The product is intended to be used on the skin," and, "The product is made for usage on a body"). The results indicated that the two products differed significantly regarding the technical characteristics, with higher ratings for the technical product ($M_{tech} = 5.52$; $M_{cos} = 1.80$; $t(50) = 9.88$, $p < .01$). At the same time, the cosmetic product resulted in higher ratings regarding the cosmetic characteristics ($M_{tech} = 2.33$; $M_{cos} = 6.48$; $t(50) = -13.68$, $p < .01$).

A second pretest ($N = 30$, M age $= 25.36 \pm 7.00$) was conducted to confirm the expectation that product category plays an important role for virtual influencer endorsement effectiveness. This time, respondents were first exposed to a picture of a virtual influencer and given a brief explanation of what a virtual influencer is. Next, we presented a range of different product categories (e.g., household products, fashion), including cosmetics and technical products, and asked the participants to which degree they could



Figure 2. Stimuli used in Study 2 (blurred for copyright reasons).

imagine virtual influencers as endorsers for products in those categories (0–100%). As expected, technical products got the highest rating (84.83%), while cosmetic products only received 36.23%. We interpreted this as a first sign that virtual influencer endorsement effectiveness may be dependent on the advertised product category.

Sample

We collected data online on $N = 158$ respondents (M age $= 34.84 \pm 14.26$), who were randomly assigned to the four experimental conditions. Again, participants were recruited via a university master's course in market research held in Germany. The instructions were the same as in Study 1. This time there were 16 participants distributing the Qualtrics online survey via email or social media. We again only questioned female participants to stay consistent with Study 1 and to make the results comparable. $N = 12$ respondents were excluded from the sample because they interrupted the survey or exceeded the time limit (> 25 minutes); four further people were excluded because they indicated that they knew of the influencer prior to the investigation (four from the virtual, zero from the human-influencer conditions). Chi-squared tests and ANOVAs showed that the groups did not differ significantly with respect to socioeconomic variables, technology affinity, technology aversion, and attitude toward the brand (all $ps > .42$).

Measures

Congruence was measured with three items (go well together, are compatible, are a good combination), adapted from Fleck, Korchia, and Le Roy (2012), completing the sentence, "The endorser and the advertised product …" and averaged ($\alpha = .85$). To measure expertise, we used the five items taken from Ohanian (1990) anchored by 1 (*not an expert, inexperienced, unknowledgeable, unqualified, unskilled*) and 7 (*expert, experienced, knowledgeable, qualified, skilled*) and averaged ($\alpha = .85$). Purchase intention was measured by a single item: "I'm likely to buy the advertised product." Attitude toward the ad was measured with the same items as in Study 1. We again included product involvement as a covariate since it was significantly correlated to the dependent variables "Congruence between endorser and product" ($r = .22$, $p = .01$) and "attitude to the ad" ($r = .21$, $p = .01$).

Recognition Check

As in Study 1, we added a recognition check in which respondents were asked to indicate whether they thought the influencer to be human or virtual. Again, we only included those respondents in the analysis who correctly classified the endorsers, resulting in the exclusion of $n = 6$ and a sample to be analyzed of $n = 136$ respondents.

Results

To test for hypothesis 4, we conducted a 2 (endorser type: human vs. virtual) × 2 (product category: cosmetic vs. technical product) ANCOVA with perceived congruence serving as dependent variable and product involvement as a covariate. Main effects of influencer type and product category were not significant ($ps > .56$). More interestingly, we found the proposed interaction effect between endorser type and product category, $F(1, 136) = 12.34$, $p < .01$, and so calculated the simple effects. Combining the virtual endorser with a technical product leads to higher congruence ratings than when combined with the cosmetic product ($M_{technical} = 4.31$, $SD = 1.45$; $M_{cosmetic} = 3.77$, $SD = 1.50$, $p = .05$). Thus, hypothesis 4 can be confirmed. For the human endorser, the effect is the other way around ($M_{technical} = 3.53$, $SD = 1.72$; $M_{cosmetic} = 4.89$, $SD = 1.78$, $p = .03$). The human endorser yields higher congruence ratings when combined with the cosmetic product than with the technical product. What is more, we found the virtual endorser to outperform the human endorser in combination with the technical product ($M_{hum+tech} = 3.53$, $SD = 1.72$; $M_{virt+tech} = 4.31$, $SD = 1.45$, $p = .04$). Figure 3 provides the congruence ratings in the four conditions.

Hypothesis 5 was tested by mediation analysis (Hayes 2018, model 4) with perceived congruence as (metric) independent variable, expertise as a mediator, and attitude toward the ad as dependent variable. A higher perceived congruence leads to a significantly higher perceived endorser expertise, $b = 0.21$, $p < .01$,

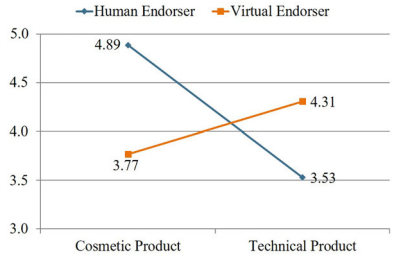


Figure 3. Congruence ratings in the four conditions.

which in turn leads to an increase in attitude toward the ad, $b = 0.38$, $p < .01$. The indirect effect was significant, $b = 0.08$, bootstrapping 95% CI [0.027; 0.157]. After mediation, there was also a direct effect of congruence on attitude toward the ad ($b = 0.19$, $p < .01$). Furthermore, we could find a significant indirect effect of congruence on purchase intention through attitude toward the ad, $b = .12$, bootstrapping 95% CI [0.067; 0.291] and in a serial mediation (Hayes 2018, model 6) through expertise and attitude toward the ad $b = 0.05$, bootstrapping 95% CI [0.015; 0.108] (see Table 3 for the results of the mediation analysis). Hence, we found support for hypothesis 5.

Discussion of Study 2

In Study 2 (hypothesis 4), we were interested in the interaction between endorser type and product category. We show that perceived congruence between endorser type (human influencer vs. virtual influencer) and the advertised product depends on the product category. The virtual endorser in combination with the technical product was perceived as more congruent than when it was combined with a cosmetic product, and subsequently outperformed the human endorser in the technical condition. Beyond that, we found a full match-up effect, also showing that congruence ratings for the human endorser were higher when combined with the cosmetic product. The virtual endorser was marked as virtually designed (and recognized as such, which was controlled for); thus, respondents might have perceived the virtual endorser differently insofar as they value the technical performance behind it and thus perceive it as more congruent with a technical than a cosmetic brand. In contrast, the "real" human endorser could be perceived as a typical and thus suitable advertising presenter for a cosmetics brand, which in turn would increase the advertising impact, encompassed by the so called "beauty match-up" hypothesis. Physically attractive endorsers should promote products that are associated with beauty (Till and Busler 2000). Koernig and Page (2002), for instance, found an attractive female endorser more effective for perfume, while a less-attractive endorser was better suited to coffee. In our case, the counterpart of the attractive human endorser was not an unattractive human, but the effect may be the same since a virtual endorser might not signal expertise for attractiveness-relevant products.

In accordance with match-up literature, congruence proved to be a central variable for ad effectiveness. A higher perceived congruence leads to higher perceived expertise and in turn to a better attitude toward the ad and purchase intention.

When taking a closer look at the results, further explanatory aspects come to mind: Not all match-up studies could find the above-mentioned match-up effects. While some studies used globally acclaimed celebrities, others relied on anonymous endorsers, which sometimes proved to be more effective (e.g., Koernig and Boyd 2009). One reason is that consumers often have strong associations with celebrities, which may play a more important role than their physical attractiveness. For example, a celebrity might be known for his or her engagement in social causes

Table 3. Effects of congruence on expertise, attitude toward the ad and purchase intention.

Mediation analysis	Unstandardized coefficient *b*	Standard error	*T*-value	*p*
Outcome: expertise				
Congruence	.211	.052	4.040	< .001
Product involvement (cov)	.007	.045	0.166	= .868
Outcome: attitude ad				
Expertise (med)	.382	.088	4.308	< .001
Congruence	.190	.056	3.356	= .001
Product involvement (cov)	.147	.042	3.437	< .001
Outcome: purchase intention				
Congruence	−.103	.101	−1.016	= .311
Expertise	−.091	.162	−.566	= .572
Attitude ad	.638	.149	4.276	< .001
Indirect effects	**Effect coefficient (partially stand. effect size)**	**Bootstrapping standard error**	**Bootstrap CI: lower level**	**Bootstrap CI: upper level**
Congruence → expertise → att. ad	.080 (.067)	.033	.027	.157
Congruence → att. ad → purchase intention	.120 (.108)	.056	.067	.291
Congruence → expertise → att. ad → purchase intention	.051 (.046)	.024	.015	.108

or for a scandalous lifestyle, and these associations come into play (even subconsciously) when evaluating congruence between endorser and product, which can be either beneficial or detrimental depending on the set of associations. When the endorsers are not known to the individual, only the available, visible cues are considered for evaluation. This reasoning indicates that the results may be different for prominent endorsers (both real and virtual).

We will review the limitations of our studies, future research potential, and implications for marketers in a general discussion.

General Discussion and Implications

The use of virtual influencers is, we believe, not only a fascinating topic, but also one of the future. While at the present time numerous brands are already turning to virtual influencers instead of traditional human ones, a further rise of their popularity is expected in the coming years. Nevertheless, there are as yet very few empirical findings on virtual-influencer assessment, making it difficult for brands to optimize their use. Our research contributes to the understanding of this new phenomenon by examining virtual influencer evaluation regarding different factors.

Our first study shows that, when advertising a cosmetic brand, a human influencer is favored over a virtual one when it comes to endorser and ad evaluation. But the virtual influencer could score in terms of perceived novelty of the ad and brand innovativeness (the question is, of course, how long this novelty factor will last). When it comes to the question of labeling, our results indicate that respondents had significant difficulty identifying the virtual influencer as such and often mistook it for a human being. Respondents also experienced a stronger feeling of uncanniness when the virtual influencer was not labeled. Thus, we advise marketers to always disclose the nature of their endorsers in order to avoid consumer confusion or the risk of deception. Recently, the Federal Trade Commission (FTC) expanded the endorser definition to include virtual influencers and to cover them under the current endorser guidelines (FTC 2022) without specifically addressing the disclosure of them being virtual. Other countries, such as India, have already issued a regulation stating that, when using virtual influencers, it must be disclosed to consumers that they are not interacting with a real human being (Lewczyk 2021). In particular, since the rapid emergence of AI and the metaverse, there has been a growing demand for specific rules for labeling virtual or AI figures to prevent consumers from feeling misled or deceived (Fitzpatrick and Rothaus 2022; Masteralexis, McKelvey, and Statz 2021). Thus, marketers should be aware that regulations on this matter could change in the near future.

Our second study shows that marketers should consider the combination of virtual influencer and endorsed product very carefully so as not generate mismatches that ruin the advertising effect. It was not known before whether the effectiveness of virtual endorsers depends on the product category. Our research fills this gap by showing that certain product categories, in our case technical products, are more congruent with virtual influencers than products emphasizing the need of a human form, such as cosmetic products. We thus answer the call of Thomas and Fowler (2021) to examine the effectiveness of AI influencers in regard to the product category and focus here on the subgroup of virtual influencers. Recent studies in the field of human endorsers highlight the relevance of a match between influencers and brands. For instance, Breves et al. (2019) showed that a strong influencer–brand fit increases attitudes via enhanced credibility. Schouten, Janssen, and Verspaget (2020) found that consumer identification with influencers is stronger than with traditional celebrities, provided consumers perceive a high similarity to the influencer. Our results extend previous match-up findings by determining that the "rules" of congruence also apply to virtual influencers as a new type of endorser. We can state that virtual influencers are not generally unsuitable as endorsers, but they need to be combined with a congruent product; then expertise is increased, and in turn attitude toward the ad.

As mentioned earlier, virtual influencers are predominantly found in the fashion and cosmetics domain. Fashionable brands do not want to miss out on this latest trend in marketing and always seek to represent the current "vibe," and may not be aware of possible counterreactions. Particularly in the beauty domain, advertisements showing ever the same flawless but generic faces are omnipresent and might have run their course. Virtual influencers could be the perfect choice to distinguish a marketing campaign from countless others. However, our results show that they are not very efficient for cosmetic products. Here, the question arises whether it might be better to present real but extraordinary human personalities rather than photoshopped, idealized, and thus interchangeable beauties.

While, at the moment, the creation and maintenance of ultra-realistic virtual characters pose major technical challenges, further advances in 3D modeling technology will most likely lead to the reduction of costs, making them attractive also for smaller businesses (Hutchinson 2019).

Managerially, we conclude that particularly brands with a technical product assortment or brands aiming to signal a special innovativeness can benefit from the use of virtual influencers. Also, product launches might profit from virtual-influencer advertising. So it is not surprising that, lately, several large tech companies have launched campaigns with virtual influencers, such as Lenovo with Imma or LG with its own virtual creation Reah Keem (Lim 2021).

Limitations and Future Research

Our research is not free from limitations. We are aware that we could not control preexisting attitudes by the design of our experiment, but we decided to present well-known brands to make the experiment as credible and realistic as possible. Other potentially relevant covariates could have an impact on attitude toward the endorser that should be considered in future studies, for example whether a person is already following the (virtual) influencer on social media. There are only a few studies examining differences resulting from follower status. Breves et al. (2021) verified that followers compared to non-followers indeed reported stronger para-social relationships and rated credibility of Instagram posts more highly. When following an influencer on social media, a consumer's emotional attachment to the influencer is stronger, and thus he or she is more inclined to accept advertising messages (Ki et al. 2020). These results would probably not be any different when examining followers vs. non-followers of virtual instead of human influencers. However, we have to ask the question, who would be interested at all in following a virtual being that does not exist in the real world? We suppose that consumers prone to the blending of fiction and reality and a sense of mystery are more likely to be in favor of virtual influencers. Moreover, the personality trait of curiosity could be a predictor for followership. Socio-demographics such as age or social class could also influence virtual influencer perception. We could not observe age-specific differences for the evaluation in both studies, but our samples in general consisted of young people.

We only focused on product–endorser fit; however, the connection between endorser and consumer may play an important role as well. Congruence between a consumer's self-concept and perception of a celebrity's personality has meaningful influence on purchase decisions (Choi and Rifon 2012). Picking up on this and on Schouten, Janssen, and Verspaget's (2020) findings that degree of identification with the endorser might differ for different endorser types, the question arises whether consumers are able to create such strong connections or even identify with virtual endorsers. Future studies should include perceived similarity or strength of identification as mediating variables in the evaluation process.

Next, we only tested two product categories, namely technical and cosmetic products. On the one hand, we need to have a more detailed look at the product categories examined, since a dishwasher, for example, can be classified as a technical or electronic product but might not be innovative per se. In the same way, a certain cosmetic product could be called innovative if it were able to clear your skin remarkably within hours. On the other hand, different product categories should be examined, such as those relating to fashion or trends in general. We further assume that especially intangible products or services could yield potential for success, such as the tourism sector. Since virtual influencers are not limited by any logistical restrictions, they can "travel" everywhere (at least in the virtual world) and thus make perfect travel endorsers. Another branch well worth considering is entertainment, especially the gaming industry. Videogame characters are based on the same technology as virtual influencers. Noonoouri, for example, is already available as a Minecraft skin, so is a playable character in the popular game. This is confirmed by our findings from the second pretest, where video games had the second highest ranking (81.83%) after technical products as potentially suitable for virtual endorsers. Conversely, a category that might be negligible is the food sector. It is to be expected that, to endorse food, an endorser needs to convey a great extent of naturalness, which virtual endorsers most likely lack.

There might be other factors underlying the product categories that could also contribute to our results. Maybe instead of the focus that a product needs a human body, it is more the "essence" of a product that might be consistent with more natural or more technical components. Or the fact that technical product functioning—at least for most people—exceeds the capacity of thought and therefore a virtual influencer itself originating from technology is a better fit. As Lil Miquela states on partnering with Samsung: "I am the product of technology myself, so I can relate to

Samsung in a way that's a lot more literal than most people."

Another limitation is that we only included female participants in our studies. Because we did not capture the male assessment, it needs to be examined whether the male perception of virtual influencers is different from the female. One variable associated with gender could be experience with video-game characters. Because males are usually fonder of video games, they might also be more familiar with the idea of virtual influencers as they can be compared to 3D gaming characters. Last but not least, culture may influence virtual-influencer perception. In countries such as Japan, individuals probably do not have any reservations about virtual influencers because virtual characters are much more prevalent than in Western countries and anime culture is ingrained into society.

Then, although we tested that the advertisements were perceived similarly, we did not use identical pictures. We tried to choose very similar expressions but still the influencers, of course, did not look the same. The selection of "existing" influencers is great for external validity, but future studies could employ exactly the same picture of an influencer and only declare that it is either human or virtual, or it could be edited in a way to make it look slightly computer animated.

There are many other aspects concerning the impact of virtual influencers that should be addressed in future studies: Do attitudes toward the influencer and the advertising change if the influencer has a "personality" that could be manipulated via a so-called human backstory? Do consumers prefer virtual influencers that can be identified as virtual on first glance, or those that look closer to humans? A lot of research still needs to be conducted to better assess the effectiveness of virtual influencers in advertising and to provide the much-needed managerial contributions for brands and marketing executives.

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